

10

STATISTICS AND PROBABILITY

ANALYSING DATA

A motor company is releasing a new model of car next year. How do they decide which colours and features to offer? They could conduct a statistical survey to determine which car features are popular.

Statistics is the branch of mathematics concerned with collecting, organising, presenting and analysing data (information). Governments, businesses, research organisations, financial institutions and spo ting groups all use statistics in their planning and decision-making.



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Chapter outline

	Working mathematically				
10.01 Interpreting graphs	U	F	PS	R	C
10.02 Misleading graphs	U	F	PS	R	C
10.03 Dot plots	U	F			C
10.04 Stem-and-leaf plots	U	F			C
10.05 The mean and mode	U	F	PS	R	C
10.06 The median and range	U	F	PS	R	C
10.07 Analysing dot plots and stem-and-leaf plots	U	F	PS	R	C
10.08 Comparing data sets	U	F	PS	R	C

Wordbank

data set A collection of data about the same subject

dot plot A special column graph of dots for a small set of numerical data

mean The average value of a data set, found by dividing the sum of values by the number of values

median The middle value of a data set when the values are arranged in order, or the average of the 2 middle values

mode The most common value(s) of a data set

outlier An extreme value that is very different from the other values in a data set

range The difference between the highest value and the lowest value in a set of data

stem-and-leaf plot A 'sideways column graph' used to list numerical data

Quiz
Wordbank 10

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (9):

10.01 Statistics
10.04 Stem-and-leaf plots 1
10.05 The mean, mode, median and range
10.06 The mean, mode, median and range • The mode, median and mean
10.07 Dot plots • Analysing dot plots 1 • Analysing dot plots 2 • Stem-and-leaf plots 2
10.08 Back-to-back stem-and-leaf plots

Twig videos (4):

10.01 Nightingale's diagram • Most popular pet
10.02 Distorted graphs: Heat wave
10.06 Average Joe

PhET interactive (1):

10.06 Centre and variability

Quizzes (5):

- Wordbank 10
- SkillCheck 10
- Mental skills 10
- Language of maths 10
- Test yourself 10

Skillsheets (3):

SkillCheck Reading linear scales
10.01 Displaying data
10.05, 10.06 Statistical measures

Worksheets (9):

10.01 Every picture tells a story • Car survey • Student survey form
10.02 Misleading graphs
10.04 Stem-and-leaf plots 1
10.06 Mean, median, mode 1
10.07 What data? • Stem-and-leaf plots 2
Mind map Analysing data

Puzzles (6):

10.01 Where all the cars are red
10.05 Looking for gold
10.06 Mean, median, mode 2 • Data puzzles • Ranges and averages
Language of maths Data crossword

Technology (3):

10.01 Creating a graph
10.06 Students' marks
10.08 Daily rainfall

Presentation (1):

10.04 Stem-and-leaf plots

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

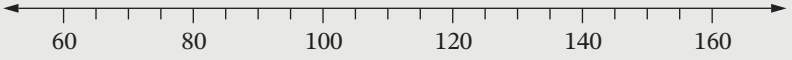
In this chapter you will:

- interpret a variety of statistical graphs, including divided bar graphs, sector graphs and line graphs
- identify graphs that are misleading
- draw stem-and-leaf plots and dot plots
- calculate the mean, mode, median and range of a set of data, including data presented on graphs and plots, and interpret these statistical measures
- compare data sets using statistics, including identifying outliers
- use primary and secondary data

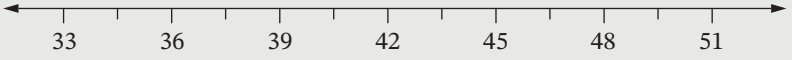
SkillCheck ANSWERS ON P. 598

1 Copy and complete each scale and state the size of one interval (unit).

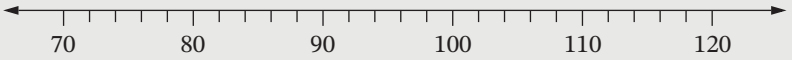
a



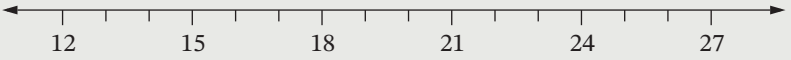
b



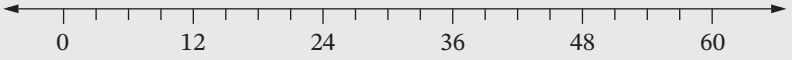
c



d



e



2 Find the average of:

a 42 and 58

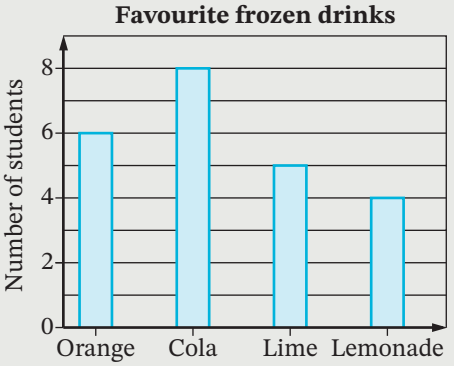
b 30 and 39

3 Some students were asked to state their favourite frozen drink. The results are displayed in this column graph.

a List the drinks in order of preference, starting with the most popular.

b How many students preferred Lime?

c How many students were in the group?



4 Write each set of numbers in ascending order.

a 31, 28, 22, 11, 24

b 21, 18, 15, 17, 13, 16

5 Write each set of numbers in descending order.

a 6, 11, 18, 17, 5

b 8, 17, 14, 23, 31, 5, 2

Quiz
SkillCheck 10

Skillsheet
Reading linear scales

Video
Statistics

10.01 Interpreting graphs

In statistics, many types of graphs are used to present **data** (information).

- **pictograms**, also called **picture graphs**
- **column graphs**, also called **bar charts**
- **divided bar graphs**
- **sector graphs**, also called **pie charts**
- **line graphs**

Skillsheet
Displaying data

Worksheets
Every picture tells a story
Car survey
Student survey form

Puzzle
Where all the cars are red

Videos
Statistics
Nightingale's diagram
Most popular pet

EXERCISE 10.01 ANSWERS ON P. 598

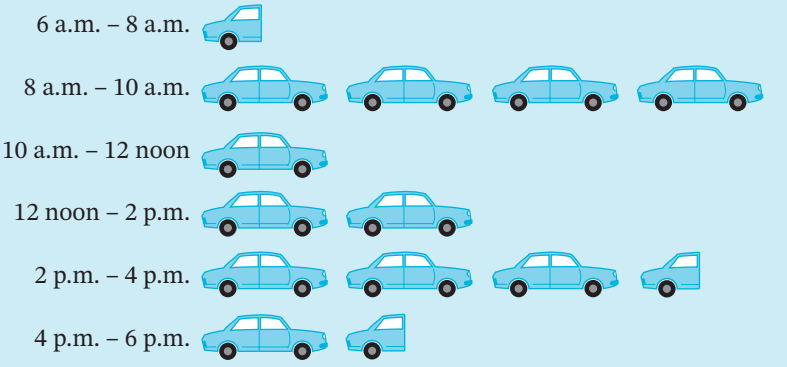
Interpreting graphs U F R C

1 **Pictograms** or **picture graphs** are used to show data about things that can be counted.

This pictogram shows the number of cars passing a school at different times during a day. R C

Key: Each  represents 10 cars.

Number of cars passing the school gate



a How many cars does each  represent?

b What does  represent? Can you see a disadvantage with this symbol?

c What is the busiest time of day for traffic?

d What is the quietest time?

e List each time period and the number of cars at that time.

f Suggest possible reasons for the flow of traffic at:

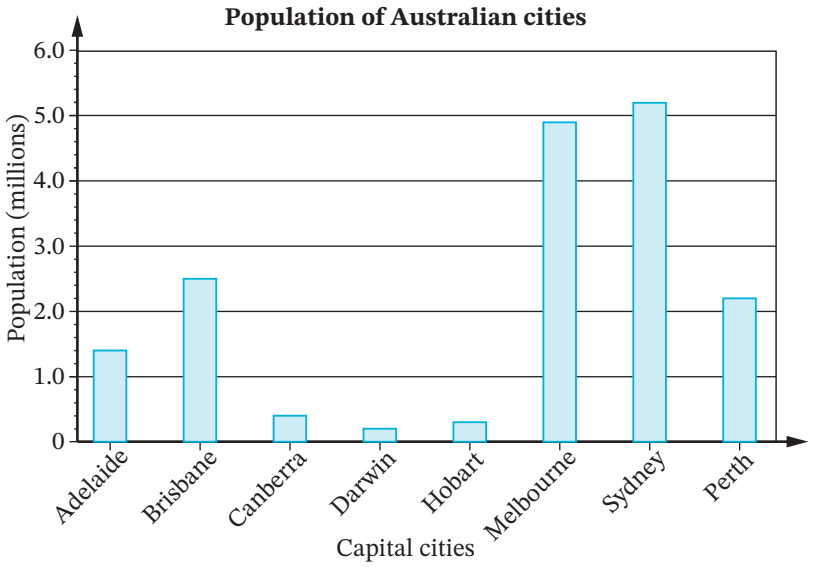
i 8 a.m. – 10 a.m. **ii** 6 a.m. – 8 a.m. **iii** 2 p.m. – 4 p.m.

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2 **Column graphs** or **bar charts** are mostly used for data that are in categories.

This column graph shows the populations of the 8 Australian capital cities. R C



a Which city has the biggest population?

b Which city has the smallest population?

c What does one interval on the vertical axis (the 'Population' axis) represent? Select the correct answer **A**, **B**, **C** or **D**.

A 0.2 people

B 1 person

C 0.2 million people

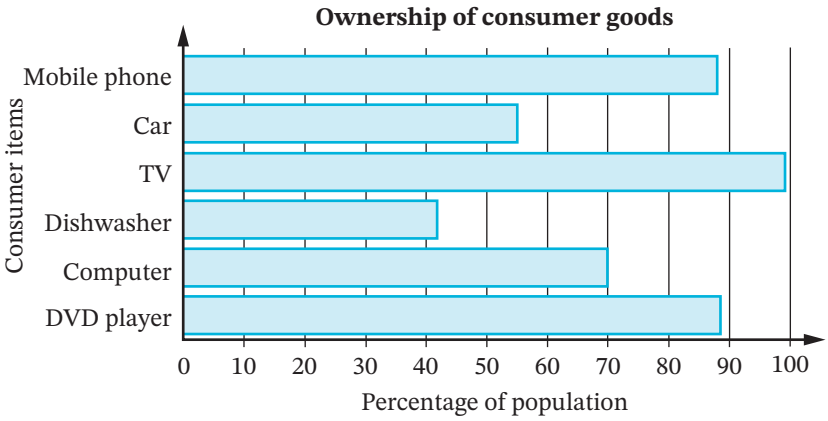
D 1 million people

d What is the population of Brisbane?

e Which city has a population of 2.2 million?

f How many times Hobart's population is Melbourne's population?

3 Sometimes a column graph is presented sideways. This graph shows the percentage of people that owned various consumer items 10 years ago. R C



a What percentage of the population owned:

i a TV? **ii** a mobile phone? **iii** a DVD player?

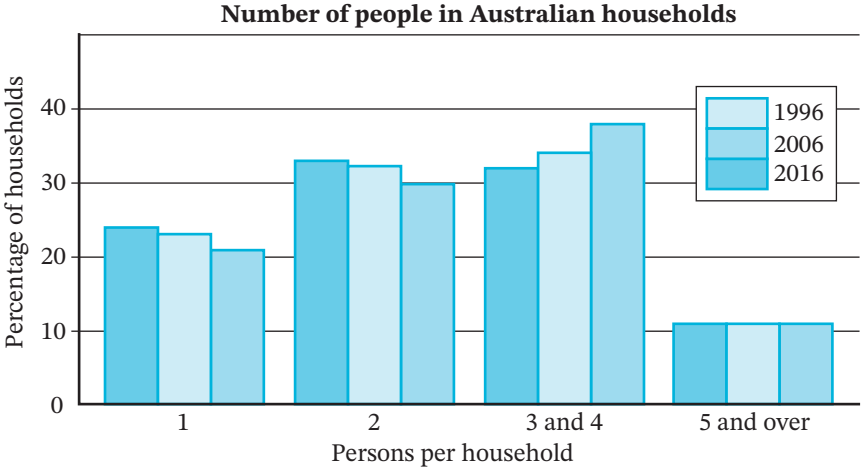
b What item was owned by 42% of the population?

c Did more people own mobile phones or computers?

d What was the percentage difference between people owning a car and people owning a computer?

e How might this column graph be different if it described the ownership of consumer items *this* year?

4 This **clustered column graph** compares the number of people living in Australian households in 1996, 2006 and 2016. R C



a What scale is used on the vertical axis?

b What percentage of households had more than 2 persons in 1996?

c In which year were 30% of households made up of 2 people?

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d Generally, which category of persons per household was the:

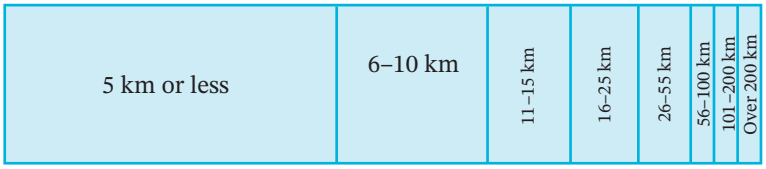
i most common? **ii** least common?

e Which category had the greatest difference between 2006 and 2016?

f True or false? 'There was a higher percentage of one-person households in 2006 than in 1996.'

5 **Divided bar graphs** are rectangular graphs used to compare parts of a whole. This divided bar graph shows the proportions of motor vehicle accidents at various distances from the driver's home. R C

Proportions of accidents at various distances from home



a What distances from home are accidents most likely to occur?

b What is the total length of this bar graph, in millimetres?

c What fraction of accidents happen within 5 km of home?

d Altogether, what fraction of accidents happen 15 km or less from home?

e What happens as you get farther from home? Suggest a reason why this may be so.

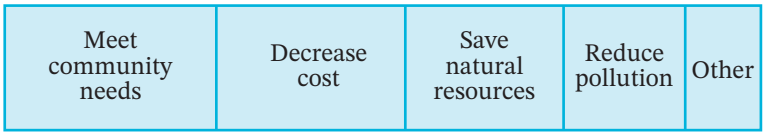
f What fraction of accidents happen more than 100 km from home?

g True or false? 'More than $\frac{3}{4}$ of all accidents occur within 25 km of home.'

h 'Smart Alec' says that to avoid having a car accident he should make all of his trips 100 km or more from home. Why is Alec wrong in saying this?

6 This divided bar graph shows the different reasons that various councils gave for having recycling programs. R C

Council's reasons for recycling



a What is the total length of this bar graph, in millimetres?

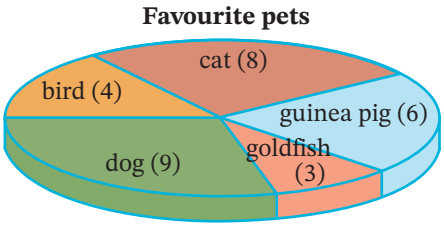
b What is the most common reason given? What fraction of the graph is this?

c What fraction of the graph represents 'Save natural resources'?

d List the reasons in order of popularity.

e What are the disadvantages of using a divided bar graph to illustrate this data?

7 Like divided bar graphs, **sector graphs** or **pie charts** are also used to compare parts of a whole. This sector graph shows the results of a class survey of 30 students regarding their favourite pet. R C



a What is the most popular pet in this survey?

b What is the least popular pet?

c What fraction of students:

i prefer goldfish **ii** prefer birds?

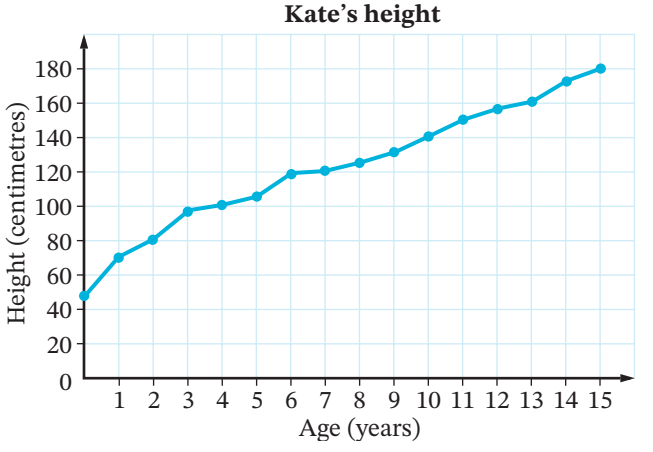
d How many degrees are there in a revolution?

e How many degrees are there in the angle of the cat sector?

f Do cats and guinea pigs together make up more than half of the preferred pets?

g If the figures from this survey are typical for all students, calculate the number of each type of pet you would expect to have if 90 students were surveyed.

8 **Line graphs** are usually drawn to show data measured over time. This line graph shows Kate's height over her first 15 years. At birth she is 48 cm tall, and at age 10 she is about 140 cm tall. R C



a What was Kate's height:

i on her first birthday? **ii** at age 7?

b At what age did Kate reach:

i 1 metre? **ii** 150 cm?

c Between which 2 birthdays did Kate grow the most? Give a possible reason.

d Between which 2 birthdays did Kate grow the least? Give a possible reason.

e How long did it take Kate to double her height from birth?

f How long did it take her to triple her height from birth? Compare this to your answer for the previous question. Why is there a difference?

g What do you think the graph will look like:

i after 15 years? **ii** after 20 years?

h How might a graph of a boy's height differ from this one?

Technology

Column graphs

Pet survey

1. Survey the students in your class on their pets. Keep a record of how many of each type of pet students have.
2. Enter your results into a spreadsheet, as shown.
3. Create a column graph. Select **Insert** and **2-D clustered column graph**. Give the graph an appropriate title and label the axes.

	A	B
1	**Pet**	**Number**
2	Dog	
3	Cat	
4	Bird	
5	Fish	
6		

Travel to school

1. Survey the students in your class on how they travel to school. Count the types of transport in your survey.
2. Enter your results into a spreadsheet, as shown.
3. Create a horizontal column graph. Select **Insert** and create a **bar chart**. Give the graph an appropriate title and label the axes.

	A	B
1	**Transport**	**Number**
2	Bus	
3	Train	
4	Walk	
5	Car	
6	Bicycle	
7		

Technology
Creating a graph

10.02

Misleading graphs 10.02

When used incorrectly, graphs can give a false or misleading impression.

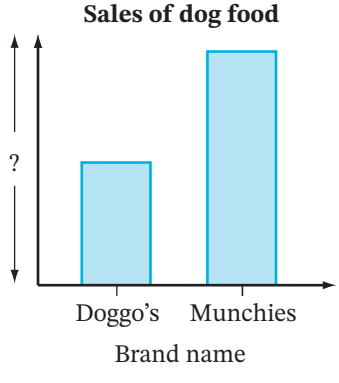
The following 3 graphs were used by *Munchies* dog food to compare its sales figures to those of *Doggo*'s dog food, but each one is misleading in some way.

Worksheet
Misleading graphs

Video
Distorted graphs: Heat wave

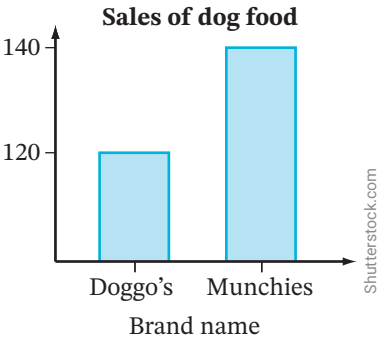
1. No scale

This graph does not have a scale on the vertical axis.
You cannot tell how big the difference is between *Munchies*' sales and *Doggo*'s sales.



2. Uneven scale

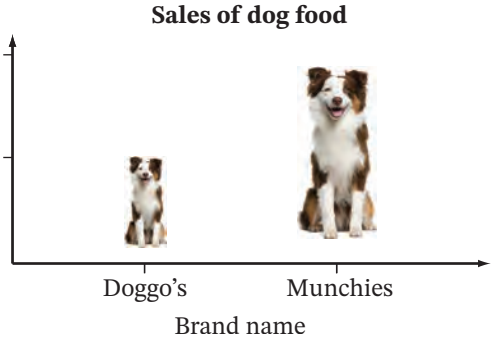
This graph shows **only part** of a scale, and the scale is not regular and does not start at 0. This makes the difference between *Munchies*' sales and *Doggo*'s sales look much greater.



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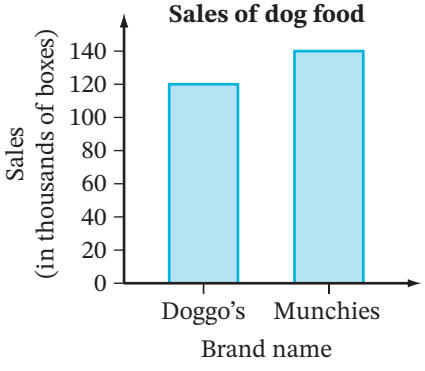
3. Incorrect use of pictures

This graph uses pictures instead of columns. The *Munchies* dog is twice as tall as the *Doggo* dog, but it is also twice as wide, making it seem much bigger (4 times as big in area).



4. A correct graph

This graph shows the information correctly. The scale is even and begins at 0. Notice that the difference in sales figures is not as large as it seemed in the 3 misleading graphs above.



Misleading graphs

A misleading graph can give a wrong impression by:

- not having a scale
- showing only part of the scale or an irregular scale
- not showing the position of zero on the scale
- using pictures instead of columns to exaggerate the differences

EXERCISE 10.02 ANSWERS ON P. 598

Misleading graphs U F R C

1 **a** What is the actual difference in sales figures between Munchies and Doggo's in the above example?

b How does the last graph illustrate this difference correctly?

c Why is it misleading to not show the position of 0 on the vertical axis?

d Why is it misleading to use pictures or diagrams on graphs instead of columns? R C

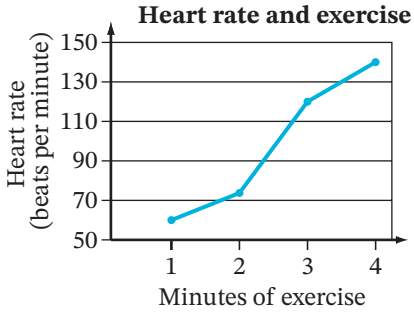
2 This line graph shows Lisa's heart rate while she is exercising on a treadmill. C

a What is Lisa's heart rate after 3 minutes?

b What is the size of one unit on the vertical axis?

c What is misleading about this graph?

d Redraw this graph correctly so that it is not misleading.



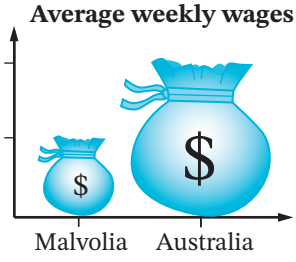
3 This graph compares the average weekly wages in Malvolia ($700) and Australia ($1400). C

a What misleading impression does this graph give?

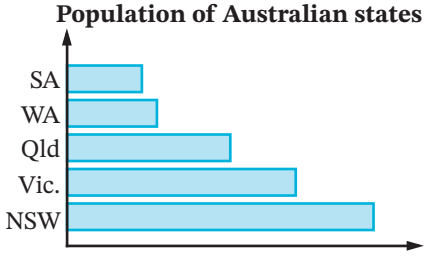
b Explain 2 things that are wrong about this graph.

c What should be drawn on the graph instead of pictures?

d Redraw this graph correctly.



4 Why is this graph difficult to interpret? C



5 **a** Draw a line graph for the profit information in this table. Place 'Year' on the horizontal axis, and use a scale of 1 cm = $2 million on the vertical axis.

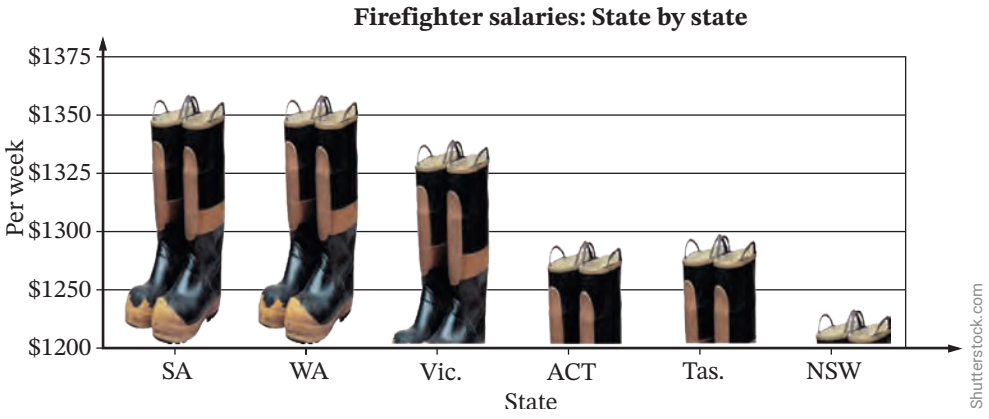
b Draw another line graph using the same information, this time using a scale of 1 cm = $0.5 million, but show only from $2 million to $4 million on it.

c Which graph looks more impressive? Why? C

Year	Profit in $ million
2016	2.5
2017	2.1
2018	3.2
2019	3.5
2020	3.6
2021	3.9

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6 This graph compares the weekly salaries of firefighters in 6 states. C



a Describe 2 ways in which this graph is misleading.

b Redraw this graph correctly.

7 Draw 2 graphs using the figures below for school sport, one that is correct and accurate, and one that gives a misleading impression. C

School sport	Hockey	Netball	Soccer
Number of players	18	20	15

Did you know?

Top 10 baby names

This table lists the 10 most popular baby names in New South Wales for boys and girls born in 2010 and 2022:

	Boys				Girls			
	2010		2022		2010		2022	
1	William	726	Noah	630	Isabella	609	Charlotte	430
2	Jack	623	Oliver	628	Chloe	604	Amelia	425
3	Oliver	558	Leo	447	Ruby	600	Isla	403
4	Joshua	549	William	427	Olivia	586	Olivia	401
5	Thomas	549	Theodore	404	Charlotte	550	Mia	363
6	Lachlan	546	Jack	381	Mia	523	Ava	343
7	Cooper	536	Henry	367	Lily	485	Ella	331
8	Noah	536	Luca	367	Emily	480	Matilda	318
9	Ethan	535	Thomas	348	Amelia	473	Sienna	293
10	Lucas	502	Charlie	334	Sienna	473	Willow	288

Source: © State of New South Wales (Customer Service NSW) CC BY 4.0 Licence (https://creativecommons.org/licenses/by/4.0/)

What names were on the list in both 2010 and 2022? What about this year?

What is the most common first name for the Year 7 students at your school?

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Dot plots

10.03

A **dot plot** is a simple type of column graph that uses dots to display the **frequency** of each data value. It is easy to draw and is useful for small sets of data. A dot plot shows:

- any gaps in the data
- any **clusters**: where values are grouped or bunched together
- any **outliers**: extremely high or low values that are very different from the other values
- how the values are spread out.

An **outlier** 'lies outside' the other values in the data set.

Outlier is pronounced 'out-ly-er'.

Example 1

The data below shows the daily maximum temperatures (in °C) in Tamworth during November.

27	28	26	25	26	27	22	30	28	29
28	26	24	22	28	24	27	29	28	27
19	25	26	29	29	26	28	28	31	25

a Construct a dot plot for the data.

b What was the highest temperature?

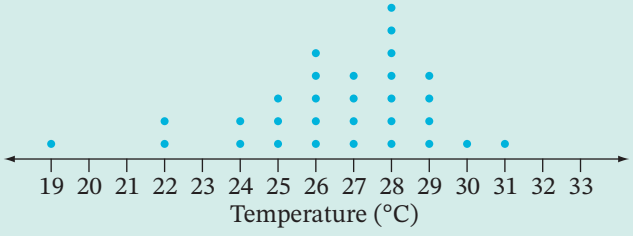
c On how many days was the temperature 25°C?

d Where were the temperatures clustered?

e What was the outlier temperature?

Solution

a



b The highest temperature was 31°C.

c The temperature was 25°C on 3 days. — 3 dots at 25

d Temperatures were clustered between 26°C and 29°C. — The most dots are bunched here

e The outlier temperature was 19°C. — Extremely low, and different from the rest of the values.

EXERCISE 10.03 ANSWERS ON P. 599

Dot plots U F R C

EXAMPLE 1

1 a The number of goals that Matias kicked in the soccer games he played last year are listed. Draw a dot plot to represent this data.

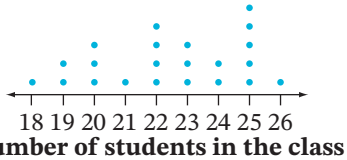
4	5	1	2	3	0	3	8	4	6	5
4	1	1	4	4	2	5	3	1	1	0

b Are there any outliers in this data? If so, what are they?

c Where is the data clustered?

d How many games did Matias play?

2 This dot plot shows the number of students in each class at Nucentry Public School. R C



Number of students in the class

a How many students are in the smallest class?

b How many classes have more than 20 students in them?

c Copy and complete: Class sizes range from _____ to _____.

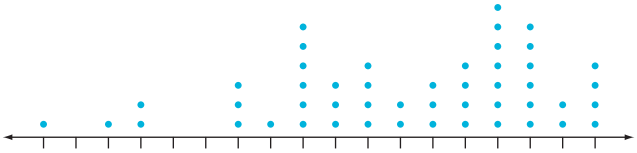
d What is the most common number of students in a class?

e Are there any outliers in this data?

f How many classes are there at Nucentry Public School?
Select the correct answer **A**, **B**, **C** or **D**.

A 8 **B** 18 **C** 22 **D** 26

3 a The ages of the people exercising at a gym one evening are shown in this dot plot. What is the most common age of people at the gym? R C



Ages of gym users

b Between which ages are the data clustered?

c Identify any outliers.

d Copy and complete: Ages range from _____ to _____.

e How many people were at the gym that evening?

4 The number of motor accidents that occurred on the motorway each day was recorded: C

1 0 0 2 0 3 5

1 0 1 0 2 3 0

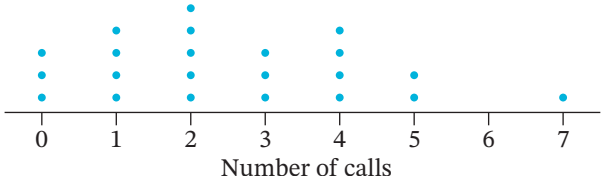
a Draw a dot plot for this data.

b What is the most common number of daily accidents?

c Calculate, correct to one decimal place, the average number of accidents per day.

d What is the outlier? Why?

5 This dot plot shows the number of phone calls made by a group of students last night.



How many students made phone calls? Select **A**, **B**, **C** or **D**.

A 7 **B** 19 **C** 22 **D** 8

10.03

Stem-and-leaf plots 10.04

A **stem-and-leaf plot** is like a sideways column graph but one that lists the actual data values on the horizontal columns. An example is shown, with the values listed in the leaf column.

Stem	Leaf
5	3 4 7 7 8 9
6	1 1 1 4 8
7	4
8	0 1 (3) 4
9	2 5 7 8

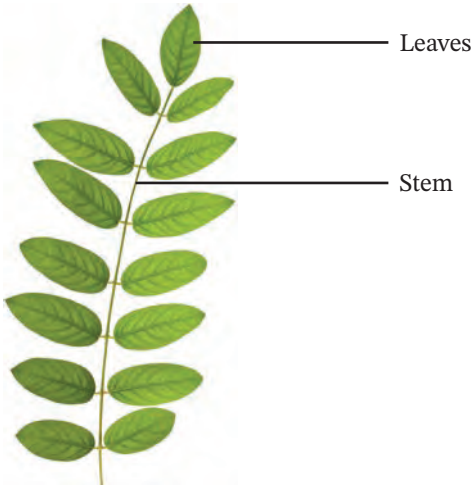
This leaf means 83.

Worksheet
Stem-and-leaf plots 1

The name 'stem-and-leaf' comes from the structure of a plant, where the stem is a branch or a vine on which the leaves hang.

An **ordered** stem-and-leaf plot shows:

- all the values, listed from smallest to largest
- the most common values
- any clusters or outliers
- how the data are spread out



Alamy Stock Photo/Y H Lim

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Video
Stem-and-leaf plots 1

Presentation
Stem-and-leaf plots

Example 2

Use a stem-and-leaf plot to show the following masses (in grams) of 30 avocados.

85	130	150	137	95	85	142	113	98	103
128	128	105	118	174	113	132	150	137	123
113	137	98	140	115	125	130	162	123	140

Solution

The masses range from 85 to 174.

We write 8, 9, 10 up to 17 down the first column to make the stem.

The leaves are the single digits written next to the stem, for example:

- 85 is shown by writing a 5 next to the 8 stem.
- 130 is shown by writing a 0 next to the 13 stem.
- 150 is shown by writing a 0 next to the 15 stem.
- 137 is shown by writing a 7 next to the 13 stem.

Stem	Leaf
8	5
9	
10	
11	
12	
13	0 7
14	
15	0
16	
17	

Continue in the same manner until all the data has been entered.

Stem	Leaf
8	5 5
9	5 8 8
10	3 5
11	3 8 3 3 5
12	8 8 3 5 3
13	0 7 2 7 7 0
14	2 0 0
15	0 0
16	2
17	4

It is more useful to rearrange the values in ascending order.

This results in an **ordered** stem-and-leaf plot.

Stem	Leaf
8	5 5
9	5 8 8
10	3 5
11	3 3 3 5 8
12	3 3 5 8 8
13	0 2 7 7 7
14	0 0 2
15	0 0
16	2
17	4

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EXERCISE 10.04 ANSWERS ON P. 599

Stem-and-leaf plots UFRC

EXAMPLE 2

1 This stem-and-leaf plot shows the heights, in centimetres, of 22 women at a gym.

What is the most common height? Select the correct answer **A**, **B**, **C** or **D**.

Stem	Leaf
14	7 8 8
15	0 1 2 3 6 6 8
16	1 5 8 9 9 9
17	2 3 3 5
18	1 1

A 15 cm **B** 156 cm

C 169 cm **D** 181 cm

2 The daily number of students served at the school canteen over a 3-week period was: C

105 76 97 88 114 86 124 101

112 98 95 105 117 81 112

a Show this information in an ordered stem-and-leaf plot.

b What were the lowest and highest numbers of students served over the period?

c On how many days were over 100 students served?

d Find the middle value. Select **A**, **B**, **C** or **D**.

A 95 **B** 101 **C** 105 **D** 112

3 The point scores of all AFL teams in their first 4 rounds of the season are shown in this stem-and-leaf plot. R C

Stem	Leaf
6	4 4 6 6 6 8
7	0 0 2 2 3 3 7 8 8 9
8	0 2 3 4 4 6 9 9
9	3 4 4 4 5 7 7 8 8 9 9 9 9
10	0 3 4 8 9
11	0 1 1 2 3 4 8 9
12	0 7
13	4 4 5 5 7
14	2 3
15	0 0 2 3 4

a How many scores are there altogether?

b How many scores are below 100 points?

c What was the most frequent score?

d Copy and complete: The team scores range from _____ to _____.

e Which stem had the most scores?

f Where are the scores clustered?

10.04

4 The heights (in cm) of the students in a PE class are:

155	153	157	166	163	162	154	175
159	157	137	162	171	140	145	168
158	141	170	166	143	175	157	177

a Make a stem-and-leaf plot of these heights.

b What are the shortest and tallest heights?

c How many students were under 150 cm tall?

d What was the most common height?

e What were the 2 middle heights?

f What fraction of the class had heights in the 160s?

g What percentage of the class had heights in the 170s? Answer correct to one decimal place.

Technology

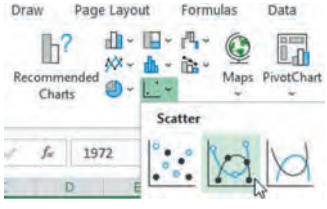
Olympic winning times

This table shows the gold medal winning times of the women's 400 m track event for the Olympic Games from 1976 to 2020.

Year	1976	1980	1984	1988	1992	1996	2000	2004	2008	2012	2016	2020
Time (s)	49.28	48.88	48.83	48.65	48.83	48.25	49.11	49.41	49.62	49.56	49.44	48.36

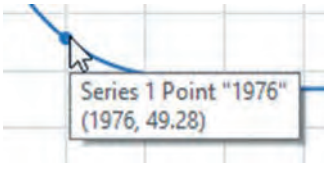
1 Enter the above data into a spreadsheet as one long table (rows 1 and 2, columns A to M).

2 To create a scatter graph choose **Insert** and **Scatter (with smooth lines and markers)**.



3 Give the graph an appropriate title and axes labels. Save the file.

4 On the graph, position the mouse over a data point. (Do not click on it.) You can view the specific details of the Olympic year and winning time. For example (see below), the winning time was 49.28 seconds in 1976.



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5 Use your spreadsheet to answer these questions.

- **a** In what year was the fastest gold medal winning time run? In cell A5, enter **=min(B2:M2)**. In cell B5, enter the year that corresponds to this time.
- **b** In cell A6, type the label 'Average'. In cell B6, use the formula **=average(B2:M2)** to calculate the average winning time for this event, from 1976 to 2020.
- **c** In cell A7, enter **=max(B2:M2)** to find the slowest winning time in this event. In cell B7, enter the year that corresponds to this time.
- **d** In cell A8, enter a formula to find the difference between the fastest and slowest winning times.
- **e** Predict the gold medal time at the 2024 Olympic Games for this event. Justify your answer, and then research the actual time.
- **f** In cell A9, enter a formula to calculate the speed, in metres per second, of the fastest women's 400 m runner, from 1976 to 2020.
- **g** Starting in cell A10, write a paragraph describing the changes in winning times for this event between 1976 and 2020.
- **h** In cell A15, suggest reasons why the pattern of gold medal times has changed between 1976 and 2020.

Mental skills 10: Maths without calculators ANSWERS ON P. 600

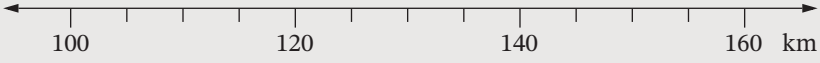
Quiz
Mental skills 10

Reading linear scales

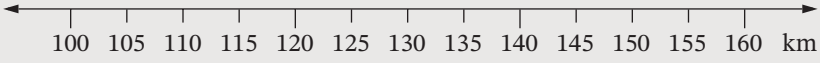
Understanding and reading the scale on a measuring instrument, on a number line or on the axis of a graph is an important mathematical skill.

1 Study each example.

a Complete the missing values on this scale.

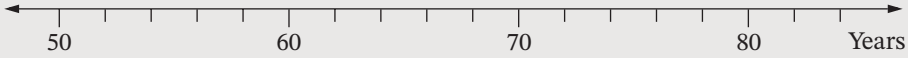


- First, choose 2 values on the scale, say 100 and 120.
- Count the number of intervals ('spaces') between the 2 values. There are 4 intervals between 100 and 120.
- To find the size of each interval, divide the difference between the 2 values by the number of intervals:
- Difference = 120 – 100 = 20 km
- Number of intervals = 4
- Size of an interval = 20 ÷ 4 = 5 km
- Use the calculated size of an interval to complete the missing values.

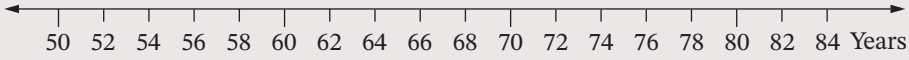


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b Complete the values on this scale.

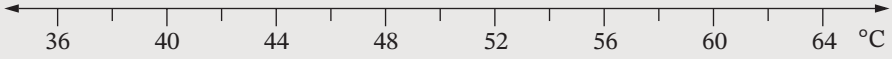


- Choose 50 and 60 on the scale.
- Number of intervals (between 50 and 60) = 5
- Difference (between 50 and 60) = 60 – 50 = 10 years
- Size of an interval = 10 ÷ 5 = 2 years.

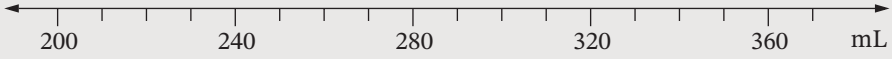


2 Now copy and complete the following scales.

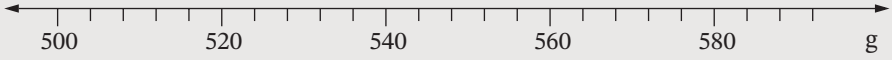
a



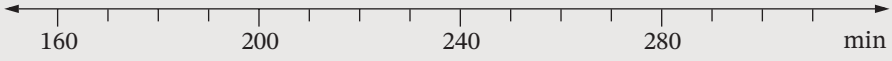
b



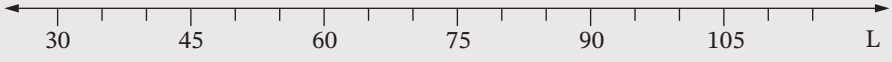
c



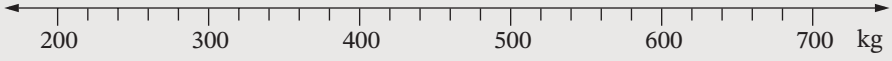
d



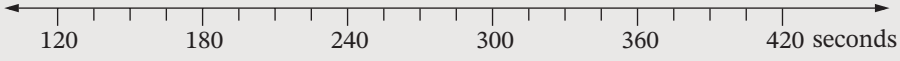
e



f



g



h



10.05 The mean and mode

Skillsheet
Statistical measures

Puzzle
Looking for gold

How do you use the word **average**?

- 'That was an average film'
- 'The average person in the street thinks ...'
- 'My average score this year is ...'

In statistics, we use the word 'average' to mean a **typical or central value** of a set of data.

The best-known average is the **mean**. The mean is found by adding all the values and dividing by the number of values.

The mean

The symbol for the **mean** is $\bar{x}$.

$$\bar{x} = \frac{\text{sum of values}}{\text{number of values}}$$

The **mode** is the most common or frequent value (or values). A set of data may have more than one mode, or no mode at all.

The mode

The **mode** is the value (or values) that occurs the most often; the value with the highest frequency.

Think: mode = '**m**ost **o**ften'

10.05



Video
The mean, mode, median and range

Example 3

The residents in a street were surveyed about the number of children living in each household. The results were:

2 2 1 2 0 3

2 1 1 4 1 0

a Find the mean, correct to one decimal place.

b Find the mode.

Solution

a Sum of values $= 2 + 2 + 1 + 2 + 0 + 3 + 2 + 1 + 1 + 4 + 1 + 0$

$= 19$

Number of values $= 12$

Mean: $\bar{x} = \frac{\text{sum of values}}{\text{number of values}}$

$= \frac{19}{12}$

$= 1.58333\ldots$

≈ 1.6 (rounding to one decimal place)

Note that the value of the mean is at the centre of the values.

The mean is about 1.6.

b The modes are 1 and 2. Both occur the most often (with a frequency of 4 each).

Example 4

Alex's scores in 8 games of ten-pin bowling were:

88 149 153 147 156 168 135 122

a Find the mean.

b Find the mode.

Solution

a Sum of scores $= 88 + 149 + 153 + 147 + 156 + 168 + 135 + 122$

$= 1118$

$$\bar{x} = \frac{\text{sum of values}}{\text{number of values}}$$

$$= \frac{1118}{8}$$

$= 139.75$

The mean is 139.75.

b There is no mode, because every score occurs the same number of times (once).

EXERCISE 10.05 ANSWERS ON P. 600

The mean and mode U F PS R C

EXAMPLE 3

1 A group of 8 children were surveyed about the amount of pocket money (in dollars) that they received each week. The results were:

20 32 32 40 18 32 18 50

a Find the mean of this set of data.

b Find the mode.

2 For each set of data, find:

i the mean (rounded to 2 decimal places, if needed)

ii the mode(s).

a	1	2	3	3	5	3	2	3	1
b	6	9	2	1	2	9	2		
c	67	43	89	65	54	86	45	76	53
d	45.1	45.0	45.4	45.1	45.8	44.6			
e	3	3	4	5	5	6	7	9	10

3 **a** Find the mode of this set of data:

blue, green, yellow, green, blue, red, green, yellow, red, green, red, blue

b Why is it not possible to find the *mean* of this set of data? C

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EXAMPLE 4

4 In a gymnastics competition, the judges awarded the following scores out of 10:

7.0 6.1 8.2 8.8 6.1 9.7 6.1 8.8

a Calculate the mean of these scores.

b Find the mode.

c Which measure (mean or mode) describes this set of scores better? Give a reason for your answer. C

5 The ages of the members of the Phuong family are:

19 31 21 3 6 14 19 24 11

The ages of the members of the Arteri family are:

19 31 21 3 6 14 19 24 91 R C

a What is the only difference between these 2 sets of data?

b Which family should have a higher mean age?

c Find the mean age (to one decimal place) for each family.

d What effect does the difference identified in part **a** have on the means?

6 A group of 6 students was surveyed on the number of phones owned in their households. The results were:

3 2 4 3 2 ☐

where ☐ represents a missing value.

What is the value of ☐ if the mean of the results is 2.5? R

Select the correct answer **A**, **B**, **C** or **D**.

A 1 **B** 2 **C** 2.5 **D** 3

7 Tanika scored 68%, 73%, 80% and 75% in 4 maths tests. R

a Calculate her mean maths mark.

b Find how much Tanika needs to score in her next maths test to increase her mean to 75%.

8 **a** Find 5 data values that have a mean of 7 and a mode of 4.

b Find 8 data values that have a mean of 10 and a mode of 12. R

9 Jamie, Sam, Karly and Tess all work at the shopping centre on Saturdays. Jamie earns $48 while Tess earns $90. If the mean of the 4 wages is $75, find possible values for Sam's and Karly's wages. PS R

10 The mean point score of a basketball team for the 30 games they played during the season is 85. What is the total number of points the team scored for the season? R

Investigation

Finding the middle data values

1 **a** Arrange these values in ascending order:

5 4 3 8 7 1 7

b Cross out the first and last values from your sorted list.

c Cross out the second and second-last values.

d Keep crossing out pairs of values from both ends until you find the middle value.

2 **a** Arrange these values in ascending order:

10 3 6 2 6 8 2 10 9 8

b Keep crossing out pairs of values at both ends of your sorted list until you find the 2 middle values.

3 **a** Arrange these values in ascending order:

9 5 5 10 4 6 6 3

b Are there one or 2 middle values? How can you tell?

c Find the middle value(s).

4 **a** Does this set of data have one or 2 middle values? How can you tell?

8 11 15 18 20 24 27 39 44

b Find the middle value(s).

5 If a set of data has an odd number of values, how many middle values does it have?

6 **a** How many values are there in this sorted data set?

2 5 9 12 17 18 27 35 39 41 45

b Is your answer to part **a** an odd number or an even number?

c What number is half of the number of values?

d Is the middle value the 5th, 6th or 7th value?

10.06 The median and range

Like the mean and the mode, the **median** is a measure of the centre of a set of data. It is the middle data value or the average of the 2 middle values.

Videos
The mean, mode, median and range
The mode, median and mean
Average Joe

The median

When data values are ordered, the **median** is:

- the middle value if there is an odd number of values
- the average of the 2 middle values if there is an even number of values.

Think: 'Median' sounds like 'medium', which is halfway between small and large.

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The range

The **range** is a measure of the spread of a set of data, and is simply the difference between the highest and lowest data values.

The range

Range = highest value – lowest value

Example 5

Tahir scored the following number of runs in a series of cricket matches:

35 98 17 54 2 22 51 45 86

Find:

a the median **b** the range.

Solution

a First, rewrite the values in order.

2 17 22 35 (45) 51 54 86 98

4 values 4 values

Median

Note that the value of the median is at the centre of the scores.

The median of the data is 45.

b Range = highest value – lowest value

= 98 – 2

= 96

Example 6

The lap times (in seconds) of 6 cyclists were:

13.5 23.1 10.2 18.4 11.9 9.3

Find:

a the median **b** the range

Solution

a Rewrite the times in order.

9.3 10.2 (11.9) (13.5) 18.4 23.1

3 values 3 values

2 middle values

$$\text{Median} = \frac{11.9 + 13.5}{2}$$

$$= 12.7$$

12.7 is halfway between 11.9 and 13.5

b Range = highest value – lowest value

= 23.1 – 9.3

= 13.8

EXERCISE 10.06 ANSWERS ON P. 600

The median and the range U F R C

EXAMPLE 5

1 For each set of data, find:

i the median **ii** the range.

a 23 20 25 22 20 21 22

b 5.5 4.5 3.4 5.3 4.9

c 7 8 3 6 5 3 5 5 4

2 The ages of the members of the Carrozza family are:

7 10 12 42 47

The ages of the members of the Binns family are:

7 10 12 38 47 R C

a What is the only difference between these 2 sets of data?

b Find the median age for each family.

c Find the range of the ages for each family.

d What effect does the difference identified in part **a** have on the medians and ranges?

3 The favourite party food for a group of 3-year-old children was recorded:

popcorn fruit chocolate fruit popcorn

chocolate chocolate fruit fruit fruit

What is the only statistical measure that can be found for this data?
Select the correct answer **A**, **B**, **C** or **D**. R

A mean **B** median **C** mode **D** range

4 Which of the following is the median of the scores below? Select **A**, **B**, **C** or **D**.

9 2 4 9 5 3 10

A 9 **B** 6 **C** 5 **D** 4

EXAMPLE 6

5 For each set of data, find:

i the median **ii** the range

a 10 8 6 4

b 36 40 38 37 40 30

c 12 13 11 14 10 15 11 12

6 11 houses were sold in Keswick Street. The selling prices are listed below:

$620 000 $625 000 $700 500 $738 000 $625 000

$1 800 000 $598 000 $612 000 $696 500 $720 000

$705 000 R C

a Find the median price.

b Find the range.



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c Calculate the mean price, correct to the nearest dollar.

d Find the mode.

e Which measure (mean, mode or median) best describes this set of house prices? Give a reason for your answer.

7 Alf's golf scores were (in order):

75 75 75 75 76 76 76 77 77 77

Mike's golf scores were (in order):

73 73 74 75 75 76 76 77 79 79 R C

a Calculate the mean score for each golfer.

b By just looking at the scores, which golfer has the higher range of scores?

c Calculate the range for each golfer.

d Who is the more talented golfer? Explain your answer.

e Who is the more consistent golfer? Explain your answer.

f Find the median score for each golfer.

8 8 friends counted the number of letters in their surnames. The results were:

4 6 7 5 4 6 9 ☐

where ☐ represents a missing value.

Find a possible value of ☐ if the median is 6 and the range is 5. R

9 **a** Find 5 data values that have a median of 7 and a range of 16.

b Find 8 data values that have a median of 12 and a range of 9. R

Investigation

Michael's family

Michael's family decided to have a family photo taken and to record the ages and heights of everyone at that time.

	Age	Height
Father	35	177 cm
Mother	33	170 cm
Big brother	11	150 cm
Big sister	10	145 cm
Michael	6	131 cm
Little brother	4	118 cm



Dreamstime.com/Famveldman

1 Find the mean age of Michael's whole family.

2 Find the mean age of the children in Michael's family.

3 What is the median age of Michael's whole family? Who is closest to this age?

4 What is the median age of the children in Michael's family? Who is closest to this age?

5 Is the mean or the median affected more when the parents' ages are not counted? What is the difference in each case?

6 Predict what would happen to the mean age if Grandpa (aged 75) came to live with the family. Test your prediction.

7 What was the family's mean age 2 years ago? How does this compare to the family's mean age now?

8 Compare the mean height with the median height of the whole family. Are there any outliers?

9 Compare the mean height and the median height of the children in the family. Are there any outliers?

10 Why is the mode not useful in this case?

11 Cousin Lee has come to stay with the family, and the mean height is now 148 cm. What is Lee's height?

Technology
Students' marks

Technology

Mean, median, mode and range

A spreadsheet can be used to calculate the mean, median and mode of a set of data.

1 Enter into a spreadsheet the following data about the daily maximum temperatures in Alice Springs in one week.

	A	B	C	D	E
1	**Day**	**Temperature (°C)**			
2	Sunday	29		Mean	
3	Monday	31		Mode	
4	Tuesday	30		Median	
5	Wednesday	33			
6	Thursday	29		Maximum	
7	Friday	28		Minimum	
8	Saturday	35		Range	
9					

2 Copy each formula into the given cells.

Cell E2: =average(B2:B8)

Cell E3: =mode(B2:B8)

Cell E4: =median(B2:B8)

Cell E6: =max(B2:B8)

Cell E7: =min(B2:B8)

Cell E8: =E6–E7

Sometimes if you type the first couple of letters, the spreadsheet will suggest the correct word.

3 Find the same data for the place where you live and enter them into your spreadsheet. Go to the Bureau of Meteorology website www.bom.gov.au to find the data.

4 Survey the students in your class and collect data such as:

- height
- number of hours slept last night
- number of children in family
- number of letters in surname

Use the spreadsheet to calculate the mean, median, mode and range for each set of data.

5 Analyse the data set from your survey.

a Are there any outliers?

b Is the data clustered around specific values?

c What other conclusions can you make for each set of data?

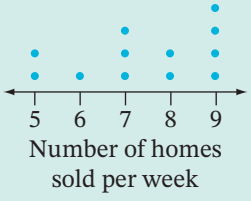
Analysing dot plots and stem-and-leaf plots

10.07

The mean, mode, median and range can be found from data displayed on dot plots and stem-and-leaf plots.

Example 7

This dot plot shows the number of homes sold per week by a real estate agency over 12 weeks.



Find:

a the range

b the mode

c the median

d the mean, correct to one decimal place.

Videos
Dot plots
Analysing dot plots 1
Analysing dot plots 2

Worksheet
What data?

Solution

a Range = highest value − lowest value

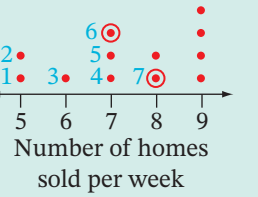
$= 9 - 5$

$= 4$

b The mode is the data value with the most dots.

Mode = 9

c There are 12 data values (12 dots). This is an even number, so there are 2 middle values (the 6th and 7th values). By counting the dots, or by crossing out pairs of dots at each end, we can see that the 6th and 7th values (circled on the right) are 7 and 8 respectively.



Number of homes sold per week

$\text{Median} = \frac{7+8}{2}$

$= 7.5$

d Mean: $\bar{x} = \frac{\text{sum of values}}{\text{number of values}}$

$= \frac{2\times5+6+3\times7+2\times8+4\times9}{12}$

$= \frac{10+6+21+16+36}{12}$

$= \frac{89}{12}$

$= 7.41666\ldots$

≈ 7.4

Note that the mean and the median are close to each other and at the centre of the data.

The mean is approximately 7.4.

Video
Stem-and-leaf plots 2

Worksheet
Stem-and-leaf plots 2

Example 8

This stem-and-leaf plot shows the number of people joining the MyFace website each day over 20 days.

Stem	Leaf
5	3 4 7 7 8 9
6	1 1 1 4 8
7	4
8	0 1 3 4
9	2 5 7 8

Find:

a the range **b** the mode **c** the median

d the mean, correct to one decimal place.

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Solution

a Range = highest – lowest
= 98 – 53
= 45

b Mode = 61

The most frequent value

Stem	Leaf
5	3 4 7 7 8 9
6	1 1 1 ④ ⑧
7	4
8	0 1 3 4
9	2 5 7 8

c There are 20 values, so the median is between the 10th and 11th values (64 and 68).

$$\text{Median} = \frac{64+68}{2}$$
$$= 66$$

d Mean: $\bar{x} = \frac{\text{sum of values}}{\text{number of values}}$

$$= \frac{53+54+57+57+\cdots+95+97+98}{20}$$
$$= \frac{1437}{20}$$
$$= 71.85$$

Note that the mean and the median are close to each other and at the centre of the data.

The mean is 71.85.

10.07

EXERCISE 10.07 ANSWERS ON P. 600

Analysing dot plots and stem-and-leaf plots U F C

In this exercise, round mean values to one decimal place where necessary.

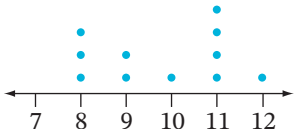
EXAMPLE 7

1 For each dot plot, find:

i the range **ii** the mode

iii the median **iv** the mean

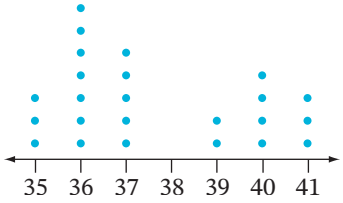
a



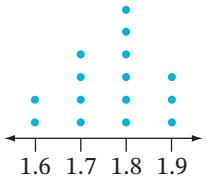
b



c



d



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EXAMPLE 8

2 For each stem-and-leaf plot below, find:

i the range **ii** the mode

iii the median **iv** the mean

a

Stem	Leaf
1	0 2 3
2	1 4 4 5 6
3	3 3 3 7
4	1 2 3 5 9

b

Stem	Leaf
7	3 4 5 7 7
8	1 2 2 3 8 9
9	0 4 4 4 4 6 7 9

c

Stem	Leaf
10	0 1 1 2
11	7 8 8 8 9
12	3 6 6 7
13	1 1 1 1 1 8
14	0

d

Stem	Leaf
0	5 5 6
1	4 4 7 7 7 7 7
2	0 3 8 8
3	
4	9 9 9

3 The maximum daily temperatures (in °C) in Cowra over a fortnight were:

10 12 10 15 14 15 11

10 19 14 11 10 11 15

Illustrate this data on a dot plot and use it to find:

a the median **b** the mode

c the mean **d** the range

4 The quiz marks out of 10 for 2 Year 8 classes are shown below.

8 Huxley: 3 2 0 1 5 8 6 7 6 3

5 4 5 6 7 9 2 5 7

8 Crancher: 7 6 3 7 8 1 9 4 6 7 2

7 2 8 10 9 9 5 7 8 9 10

a Draw a dot plot for the data of each class.

b What is the mode of the marks for 8 Huxley?

c What is the median of the marks for 8 Crancher?

d What is the range of the marks for 8 Huxley?

e Calculate the mean for 8 Crancher.

f Which do you think is the 'better' class? Give a reason for your answer. C

5 This stem-and-leaf plot shows the number of students buying from the school canteen each day over a 3-week period.

Stem	Leaf
7	6
8	1 6 8
9	5 7 8
10	1 5 5
11	2 2 4 7
12	4

a What is the range for this data?

b Find the median.

c What are the modes?

d Calculate the mean.

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6 This stem-and-leaf plot shows the heart rates (in beats per minute) of people riding on a rollercoaster at a theme park.

Find:

a the mode **b** the range

c the median **d** the mean.

Stem	Leaf
4	0
5	2 3 5 8 9
6	0 1 1 2 4 4 6 7 7 7 8 8 9
7	0 1 3 3 7 9
8	2 5 6 7
9	1

Comparing data sets 10.08

When we analyse data, we try to describe or summarise the information. This allows us to notice patterns and trends and to draw some conclusions from them. We can use the mean, mode, median and the range to make comparisons between sets of data.

Technology
Daily rainfall

Statistical measure	Features	When is it appropriate to use?
Mean $\bar{x} = \frac{\text{sum of values}}{\text{number of values}}$.	• Depends on all the values in the data set. • Affected by outliers.	When the data set does not have many extreme values (outliers).
Mode Most common value(s)	• There may be more than one mode, or no mode at all. • Not affected by outliers.	When the most common value or category is needed.
Median Middle value, or average of the 2 middle values, when values are arranged in order	• Can be one of the values. • Not affected by outliers.	When the data set has extreme values (outliers).
Range Highest value – lowest value	• Depends on the highest and lowest values only.	When a measure of spread is needed.

The mean, mode and median are called **measures of centre** while the range is a **measure of spread**.

Example 9

The results of a survey investigating the number of boys and girls who visited a shopping centre each day for 12 days are as shown.

Boys: 105, 76, 97, 88, 114, 86, 124, 102, 111, 97, 96, 81

Girls: 78, 102, 99, 89, 113, 116, 99, 108, 98, 116, 114, 97

a Show this information on a back-to-back stem-and-leaf plot.

b Calculate the range for each set of data.

c Find the median for each set of data.

d Comment on the differences between the data for boys and girls.

Video
Back-to-back stem-and-leaf plots

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Solution

a A back-to-back stem-and-leaf plot combines 2 stem-and-leaf plots, sharing the same stem.

Boys				Girls			
		6	7	8			
8	6	1	8	9			
7	7	6	9	7	8	9	9
	5	2	10	2	8		
	4	1	11	3	4	6	6
		4	12				

b Boys: Range $= 124 - 76$
$= 48$

Girls: Range $= 116 - 78$
$= 38$

c Boys: Median $= \frac{97+97}{2}$
$= 97$

There are 12 values, so the median is the average of the 6th and 7th values.

Girls: Median $= \frac{99+102}{2}$
$= 100.5$

d The boys' data is more spread out but the median for the girls is higher.

EXERCISE 10.08 ANSWERS ON P. 601

Comparing data sets U F R C

EXAMPLE 9

1 Two brands of batteries were tested in the same toy to determine which lasted longer. The back-to-back stem-and-leaf plot shows the 2 sets of data, recorded to the nearest hour. R C

Dynamo									Energy Plus								
						4	1	1	3	3	7	7	8	8	9		
				8	8	6	3	2	5	6	9	9	9	9	9		
8	7	7	6	6	5	4	4	3	2	4	6	7	7	8	8	9	9
	6	5	5	4	3	2	2	4	0	1							
				3	2	0	0	5									

a How many batteries of each brand were tested?

b Find the mean, median and range for each set of data.

c Comment on the differences between Dynamo batteries and Energy Plus batteries.

d Which brand do you think is better? Explain your answer.

2 The 3D-TV MegaStore recorded the weekly number of sales at 2 stores over a 20-week period. R C

Hurstville: 34, 43, 45, 55, 66, 71, 78, 35, 83, 86, 94, 81, 75, 68, 66, 96, 34, 66, 71, 83

Penrith: 96, 36, 86, 81, 35, 46, 38, 33, 56, 66, 66, 48, 54, 71, 81, 37, 48, 56, 55, 40

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a Display this information on a back-to-back ordered stem-and-leaf plot.

b Calculate the mean, mode, median and range for each store.

c Which store performed better? Explain your answer.

d Which sales figure from the Penrith store is an outlier? Which of the measures calculated in part **b** are most affected by this outlier?

3 The marks out of 50 for the same English test, scored by 2 different classes of Year 7 students are listed below. R C

7 Murray marks: 35, 44, 40, 48, 47, 42, 47, 45, 38, 38, 31, 32, 38, 50, 43, 31, 49, 31, 47, 37, 48, 46, 29

7 Winton marks: 40, 41, 46, 47, 47, 36, 33, 32, 26, 39, 48, 44, 31, 35, 31, 29, 45, 48, 45, 29

a Draw a back-to-back stem-and-leaf plot for these 2 data sets.

b What is the highest mark overall?

c What is the lowest mark overall?

d In class 7 Murray where are the marks clustered?

e Copy and complete:

i In 7 Murray, the marks range from _____ to _____.

ii In 7 Winton, the marks range from _____ to _____.

f Find the mean (to one decimal place) and the median for each class.

g Which do you think is the 'better' class? Give a reason why you think this.

h Identify any outliers.

4 2 groups of students achieved these marks out of 10 for their PE project:

Group A: 5 5 5 5 6 6 6 7 7 7 8

Group B: 3 3 4 5 5 6 6 7 9 9 10 R C

a Calculate the mean (to one decimal place) and the median of each group's marks.

b Draw a dot plot for each group and describe the differences between the way each group has its marks spread out.

c Group C achieved these marks, but one student was away.

Group C: 4 5 5 6 6 7 7 7 10

What mark would the absent student need to achieve to give this group the same mean as the other 2 groups? (*Hint*: What total does each group need?)

5 This back-to-back stem-and-leaf plot shows the number of goals scored in each match by 2 basketball teams during last season. R C

Cobar Cougars								Tilba Tigers					
		6	6	5	4	3	4	4	9				
			8	8	3	0	5	2	3	3	6	8	
8	8	6	6	3	1	1	6	5	6	8	9		
			7	4	3	0	7	0	0	1	3	6	
				6	6	5	8	2	5	7	7	9	9
					2	2	9	0	3	4			

10.08

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a How many games did these teams play in one season?

b Find the mean, median, mode and range for each team.

c Comment on the differences between the Cobar Cougars and the Tilba Tigers.

d Which team do you think is better? Explain your answer.

6 A basketball coach kept a record of the number of points scored by Maria and Stacey in each match over 12 weeks. R C

Maria:	41	38	25	19	53	35	30	32	39	45	46	37
Stacey:	35	30	30	24	37	25	37	44	20	29	29	35

a Display this data on a back-to-back ordered stem-and-leaf plot.

b Calculate the mean (to one decimal place), mode, median and range for each player.

c Who do you think is the more consistent player? Explain.

d Who is the higher scorer?

e Who do you think is the better player? Why?

Power plus ANSWERS ON P. 601

1 The mean of 5 numbers is 24. If a 6th number is added, the mean of the 6 numbers is 21. Find the 6th number.

2 A small class obtained these results in an exam:

66 68 74 76 82 79

a Find the mean of these marks, correct to 2 decimal places.

b The teacher realised there was an error in the marking and added 3 to each mark. Find the mean of the new marks.

c What effect did the extra 3 marks have on the mean?

d What effect does adding or subtracting the same number for all of the data have on the mean?

3 Melissa sat 5 exams. Her average mark was 74%. What mark should Melissa obtain in the 6th exam if she wishes her average mark for the 6 exams to be 77%?

4 5 data values were collected, but the figures were lost. The mean of the data was 8 and the median was 9.

a What was the total of the data?

b What could the data have been?

c If you are now told that the range is 7, what could the set of data have been?

5 4 sisters work at the same bank. One earns $500 per week and another earns $800 per week. The mean weekly wage of the 4 sisters is $2000. Is this possible? How?

CHAPTER 10 REVIEW

Language of maths

analyse	average	centre	cluster
data	dot plot	extreme	mean
median	middle	mode	ordered
outlier	range	spread	stem-and-leaf plot

1 Look up the word 'average' in the dictionary. What does it mean?

2 What is another name for an extreme data value?

3 Why are the dot plots and stem-and-leaf plots called 'plots' rather than 'graphs'?

4 Which word is an example of a measure of **spread**?

5 Explain what is meant by a **cluster** of data values.

6 What is a 'median strip'? Why do you think it has this name?

Quiz
Language of maths 10

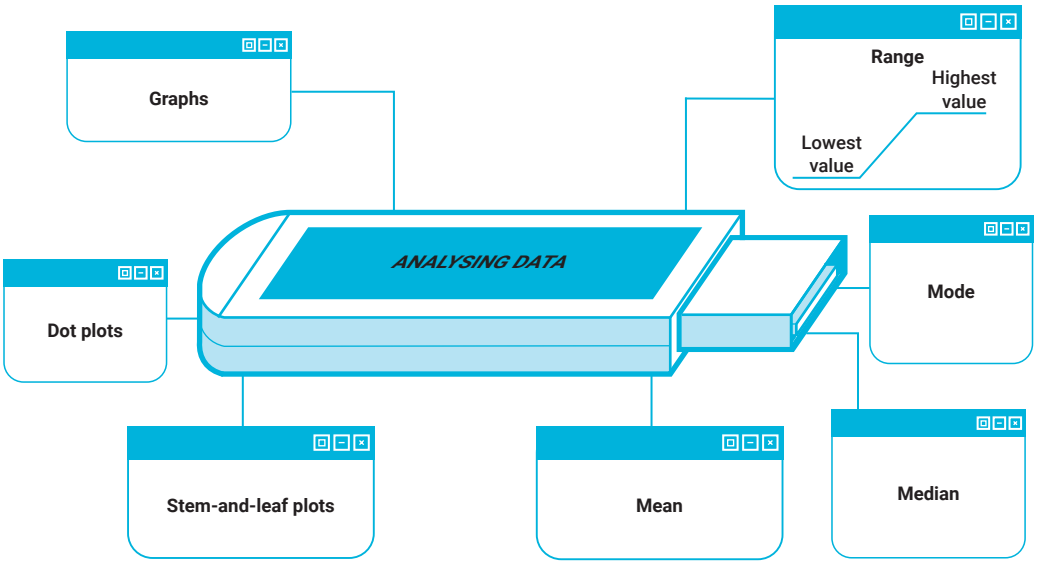
Puzzle
Data crossword

Topic summary

- Write about what you have learnt in this chapter.
- Was this work new to you? If not, in what subject have you studied it?
- Did you have any difficulties? Discuss them with a friend or your teacher.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

Worksheet
Mind map
Analysing data



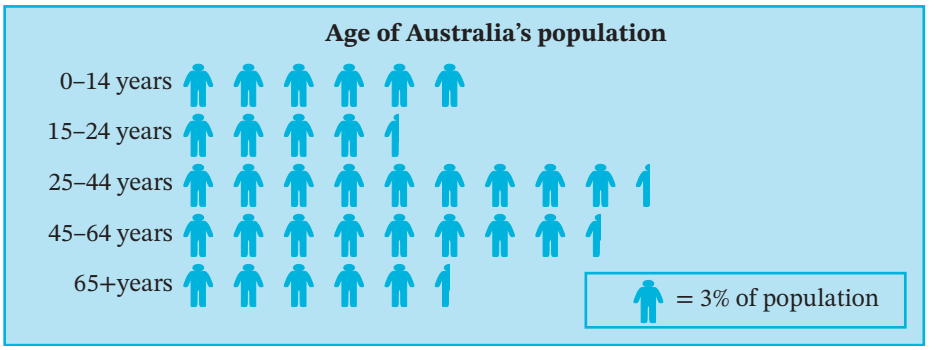
Quiz
Test yourself 10

TEST YOURSELF 10

ANSWERS ON P. 601

10.01

1 **a** What age group has the smallest percentage of the population?



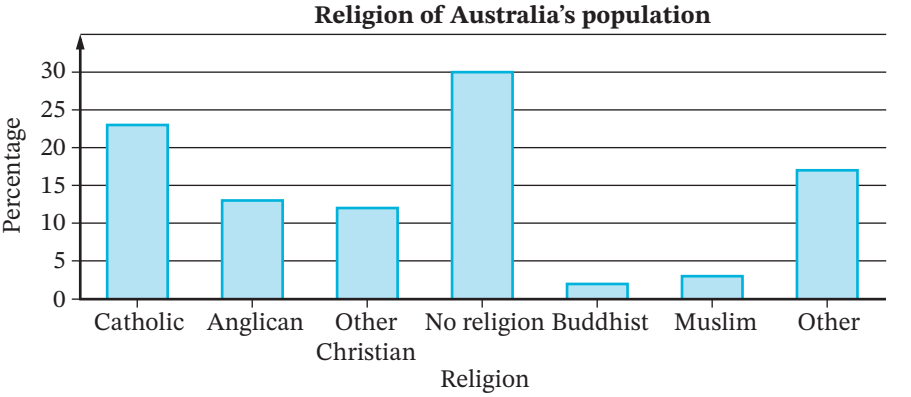
Source: Australian Bureau of Statistics CC BY 4.0 Licence (https://creativecommons.org/licenses/by/4.0/)

b What percentage are in the 15–24 years age group?

c Why might the 25–44 years group be the largest?

10.01

2



a What percentage of the population is Catholic?

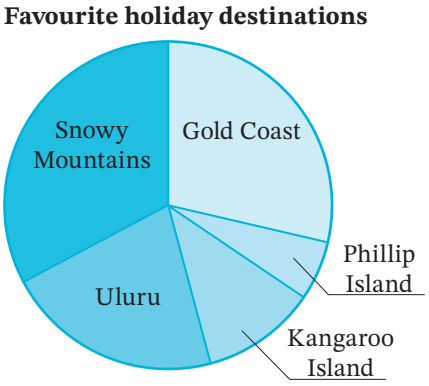
b Which religion is followed by 13% of the population?

c How might this graph have been different 20 years ago?

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3 This sector graph shows the favourite holiday destinations of 80 people surveyed at a city shopping centre. 10.01



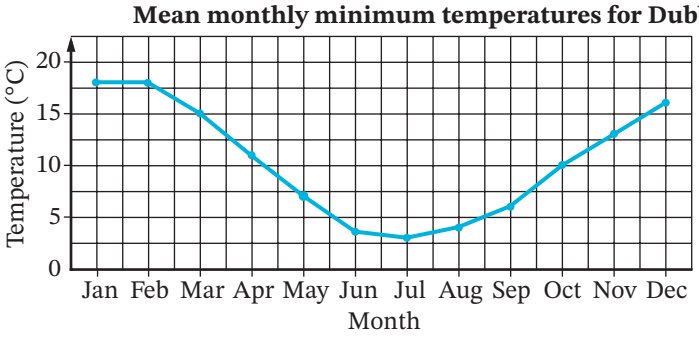
a What is the most popular destination?

b Which holiday destination was preferred by 17 people?

c Estimate how many people preferred the Gold Coast.

d True or false: Fewer than 10 people preferred Phillip Island.

4 This graph shows temperature data for Dubbo. 10.01



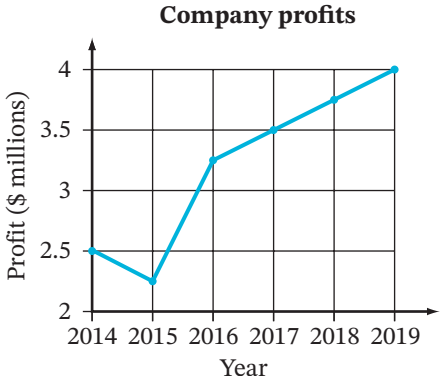
a Which months have the highest mean minimum temperature? What is that temperature?

b Name any 2 months that have the same mean minimum temperature.

c Which month has a mean minimum temperature of 10°C?

d Between which 2 consecutive months is the smallest drop in mean minimum temperature?

5 This line graph shows the yearly profits of a company over a period of 6 years. 10.02



a What is incorrect about this graph?

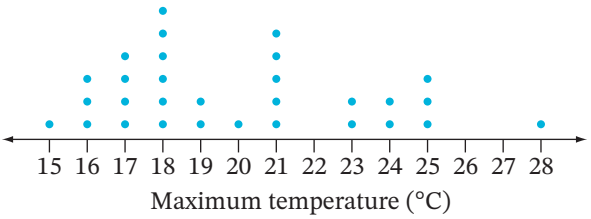
b What misleading impression does this give?

c Redraw the graph correctly.

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10.03

6 The daily maximum temperatures (in °C) at Bega during April are shown in this dot plot.



a What is the mode?

b What is the outlier?

c On how many days was the temperature 25°C?

d On what fraction of days did the temperature drop below 20°C?

10.04

7 This unordered stem-and-leaf plot shows the masses of the players of a football team, in kilograms.

Stem	Leaf
7	9 5 8 1 6 9
8	4 2 5 0 0 8 0 3 2 4 2 8
9	0 8 4 2 2 0 0 8 0 5 6
10	5 4

a How many players are on the team?

b Present this information as an ordered stem-and-leaf plot.

c Which mass occurs most often?

d What is the lowest mass?

For the rest of this exercise, round mean values to 2 decimal places where necessary.

10.05

8 For each set of data, find:

i the mean **ii** the mode.

a 4 3 2 5 6 4 4 **b** 6 12 11 12 10 6 6 10 6 **c** 8 4 1 1 4 1 3 6

10.06

9 For each set of data in question **8**, find:

i the median **ii** the range.

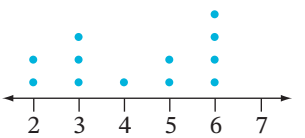
10.07

10 For each plot below, find:

i the range **ii** the median

iii the mode **iv** the mean.

a



b

Stem	Leaf
4	0 0 1 2
5	1 3 4
6	6 6 6 7 8
7	4 5

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10.08

11 The assignment marks for 20 girls and 20 boys are as follows:

Girls:	75	28	37	35	60	73	69	52	94	66
	55	39	48	51	53	18	29	76	59	83
Boys:	88	29	38	72	50	74	73	30	85	10
	28	93	66	17	75	40	55	62	73	58

a Construct a back-to-back ordered stem-and-leaf plot for the data.

b What was the highest mark? Who scored it, a boy or a girl?

c Find the mean, median, mode and range for each group.

d Comment on the differences between the girls and the boys on this assignment.

e Which group of students seemed to be more consistent in the marks they scored, the girls or the boys? Give reasons for your answer.

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